

Here is a comprehensive list of 60 questions, broken down into 20 Operational, 20 Technical, and 20 IT questions. These are designed to rigorously assess a large-scale mining company's environment, exposing the pain points your disruptive LIMS can solve.

Part 1: Operational Questions

Target Audience: Lab Managers, Senior Chemists, Metallurgical Superintendents.

Focus on physical workflows, throughput, and human factors.

1. What is your average daily sample throughput, and how much does it fluctuate during peak exploration vs. steady-state mining periods?
2. Walk me through the physical lifecycle of a sample from the moment it arrives at the lab to the moment the data is reported. Where are the manual bottlenecks?
3. How are lab technicians currently capturing data on the floor (e.g., paper logbooks, desktop terminals, ruggedized tablets)?
4. Given that technicians wear heavy PPE (gloves, face shields), how user-friendly is the current data entry interface? Do you see workarounds being used?
5. Describe your shift handover process. How long does it take, and how much critical context is lost between outgoing and incoming shifts?
6. How do you currently track the physical "genealogy" of a sample (e.g., tracking a 1kg bulk sample as it is split into 30g pulps, and eventually a gold prill)?
7. What is your standard operating procedure (SOP) when a high-grade sample result is detected? How is it secured?
8. How do you manage the physical chain of custody for doré bullion bars from the lab to the vault?
9. How are lab consumables (flux, lead, crucibles) and highly sensitive reagents (acids, cyanide) currently tracked, stored, and reordered?
10. When a QC sample (blank, duplicate, standard) fails, what is the operational protocol for halting production and re-assaying?
11. How are custom or non-routine assays (e.g., metallurgical test work vs. standard grade control) handled differently in your workflow?
12. How is communication handled between the lab and the mill/plant when incoming ore grades change drastically?
13. What is the process for disposing of hazardous waste (e.g., lead-laden cupels, cyanide tails), and how is it logged?
14. How are re-assays prioritized and scheduled without disrupting the daily influx of new production samples?

15. What manual reporting takes up the majority of the Lab Manager's time at the end of every shift or month?
16. How do you handle and track sample rejections (e.g., samples that are too wet, contaminated, or mislabeled upon receipt)?
17. What is your current average turnaround time (TAT) for a fire assay, and what steps are you taking to reduce it?
18. How do you manage and track the calibration of physical equipment (balances, furnaces, pipettes) on a daily/weekly basis?
19. How are new lab technicians trained on your current LIMS, and how long does it take for them to become proficient?
20. If a sample is physically lost or misplaced in the lab, what is the operational procedure to locate it or flag the missing data?

Part 2: Technical Questions

Target Audience: Lab Supervisors, QA/QC Managers, Analytical Chemists. Focus on instruments, assay methods, and data integrity.

1. What specific analytical instruments (make and model) are you currently running for AAS, XRF, and ICP-MS analysis?
2. What is the age of your oldest critical analytical instrument, and does it rely on legacy communication protocols (e.g., RS-232) or modern network interfaces?
3. How is data currently transferred from your analytical instruments to your LIMS? Is it fully automated bidirectionally, or does it require manual CSV exports?
4. Do your instruments support direct worklist downloading from a LIMS, or must operators manually enter sample IDs at the instrument terminal?
5. What are your standard assay methods for Gold (e.g., Pb-collection fire assay, aqua regia digest), and do you require the LIMS to calculate specific flux recipes based on ore matrix?
6. How does the current system handle limit of detection (LOD) and upper limit of measurement (ULM) flagging for high-grade or trace samples?
7. How are QC insertion rules configured? Can the system automatically blind-insert blanks and duplicates at defined frequencies, or is this done manually?
8. What statistical QC rules (e.g., Westgard rules) are applied to your control charts, and does the system auto-lock batches when failures occur?
9. How are Certified Reference Materials (CRMs) managed and tracked in the system to ensure they haven't expired or degraded?

10. How does the system handle the merging of data from multiple instruments for a single sample (e.g., Au by fire assay + Cu by XRF)?
11. What specific metadata must be captured for every sample (e.g., drill hole ID, depth, ore type, moisture content)?
12. How are instrument calibration drifts detected, logged, and corrected within the current system?
13. If an instrument crashes mid-run, how does the current software handle partial data sets and sample recovery?
14. How are wet chemistry results (e.g., cyanide titrations) integrated with fire assay results in the final database?
15. For bullion certification, how are specific gravity (density) measurements and trace element impurities (Pd, Pt, Fe, Pb) combined into a final Certificate of Analysis (CoA)?
16. Does the current system support bidirectional communication with analytical balances to auto-populate weight fields without manual typing?
17. How are historical assay results accessed and compared against new data for trend analysis?
18. What file formats are currently used to export assay data to geological modeling software (e.g., CSV, XML, ACS files)?
19. How are duplicate and pulp reject samples tracked and stored for future re-assay or metallurgical testing?
20. What specific limitations in your current system prevent you from achieving ISO 17025 compliance more easily?

Part 3: IT Questions

Target Audience: IT Directors, Enterprise Architects, System Administrators. Focus on infrastructure, security, architecture, and integrations.

1. Is your current LIMS architecture purely on-premise, cloud-hosted, or a hybrid model?
2. Given the remote locations of some sites, how stable is the network connectivity, and how does the LIMS behave when the WAN connection drops?
3. If the system operates offline, how does it handle database conflict resolution when connectivity to the central server is restored?
4. What is the underlying database architecture of your current LIMS (e.g., SQL Server, Oracle, PostgreSQL), and are there performance issues with large datasets?

5. How is user authentication managed? Are you using Active Directory / Microsoft Entra ID for Single Sign-On (SSO)?
6. How granular is the Role-Based Access Control (RBAC)? Can you restrict access down to specific screens, fields, or individual lab sites?
7. How does the LIMS integrate with your corporate ERP (e.g., SAP, Oracle) for financial accounting and inventory management?
8. Are there robust REST or GraphQL APIs available for third-party systems to pull real-time assay data, or do you rely on overnight batch jobs?
9. How are electronic signatures handled to ensure compliance with 21 CFR Part 11 (non-repudiation)?
10. What are your corporate data retention policies, and how does the LIMS handle the archiving of historical assay data?
11. What are the disaster recovery and backup strategies for the LIMS database? What is the targeted Recovery Time Objective (RTO)?
12. How are system audit logs secured? Can a database administrator alter or delete historical audit trails?
13. What cybersecurity measures are in place to protect lab endpoints (tablets, PCs) from unauthorized access or malware?
14. Are mobile devices (e.g., iOS/Android tablets) allowed on the lab network for data entry, and how are they managed (MDM)?
15. How are software updates and patches deployed to the LIMS? Does this require significant system downtime?
16. What is the typical IT support response time for lab-related system issues, and where is the support team located?
17. How is master data (e.g., new sample types, assay methods, user lists) managed and synchronized across multiple global mine sites?
18. What are the biggest IT headaches in maintaining, upgrading, and scaling the current LIMS infrastructure?
19. How open is your IT department to adopting a SaaS-based model for a critical operational system like a LIMS?
20. What security and compliance audits (e.g., SOC 2, ISO 27001) would a new SaaS vendor need to pass to be approved by your IT governance team?